DD-B-00592a(DSA-DM)
31 March 1965
Superseding
DD-B-592
30 March 1960

INTERIM FEDERAL SPECIFICATION

BOTTLE, STERILE FLUID STORAGE

This interim Federal specification was developed by the Defense Medical Supply Center, Third Avenue and 29th Street, Brooklyn, New York 11232, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification DD-B-592, dated 30 March 1960.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type and two sizes of square-shaped glass bottles, fitted with an automatic sealing closure cap, suitable for the preparation and storage of irrigating solutions under vacuum for prolonged periods.

1.2 Classification. The sterile fluid storage bottles shall be of the following sizes, as specified (see 6.1):

Size I 500 cubic centimeters capacity

Size II 1000 cubic centimeters capacity

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

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Federal Specifications:

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard

Federal Standards:

Fed. Std. No. 102 Preservation, Packaging, and Packing Levels.
Fed. Test Method Std. No. 406 Plastics: Method of Testing.
Fed. Test Method Std. No. 601 Rubber; Sampling and Testing.
Fed. Test Method Std. No. 626 Glassware; Methods of Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards, as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specification:

MIL-L-10547 Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Glass. Bottle shall be made of heat-resistant, transparent, lead-free borosilicate glass, having a maximum resistance to the corrosive action of distilled water. The glass shall have a linear coefficient of expansion of 32×10^{-7} to 41×10^{-7} per degree centigrade (C.) between 0° and 350° C., when tested as specified in 4.4.1.1. The glass shall not require more than 1.0 milliliter (ml.) of 0.02 normal sulfuric acid to neutralize alkalinity per 10 grams of glass, when tested as specified in 4.4.1.1. The glass shall be fully annealed, and shall show no ring strain when tested as specified in 4.4.2.1.1. Any longitudinal strain, if present, shall be faint and highly diffused. The bottles shall be capable of withstanding 45 pounds per square inch of bursting pressure when tested as specified in 4.4.2.1.2.

3.1.2 Plastic. The closure cap shall be made of a black phenolic-type plastic molding material, having the following physical characteristics, when tested as specified in 4.4.1.2.

Water absorption, percent	0.4 maximum
Distortion under heat—	
°Fahrenheit (F.)	320 minimum

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There shall be no appreciable changes in the above physical properties after the plastic specimens have been subjected to the deterioration tests specified in 4.4.1.2.1. The finished cap shall withstand autoclaving, as specified in 4.4.2.2, without distortion.

3.1.3 Rubber. The collar shall be made of a black, nontoxic, nonblooming, neoprene rubber material, having the following characteristics, when tested as specified in 4.4.2.3.

Durometer hardness (unaged) ---------- 48 maximum

Durometer hardness
(after autoclave aging ---------- 53 maximum

3.2 Construction. Sterile fluid storage bottles shall consist of a bottle, cap, and collar as specified herein:

3.2.1 Bottle. The bottle shall be molded of glass as specified in 3.1.1, and shall have an average thickness of 3/16 inch with no section having a thickness of less than 3/32 inch.

3.2.1.1 Size I bottle. The mouth of size I bottle shall be molded as shown in figure 1. In order to allow for evaporation during sterilization, the filling mark shall be at a volume of 550 cubic centimeters (cc.) ± 2 percent, when tested as specified in 4.4.2.1.3. All four sides of the bottle shall have molded in legible, raised letters "For External Use Only", as shown in figure 1. On two opposite sides of the bottle the capacity level shall be indicated by legible, raised horizontal lines in increments of 50 cc., from 50 to 550. At least the 100, 200, 300, 400 and 500 level lines shall be numbered. Under the 550 level line the words "Fill To Here" shall be molded in legible, raised letters.

3.2.1.2 Size II bottle. The mouth of Size II bottle shall be molded as shown in figure 2. In order to allow for evaporation during sterilization, the filling mark shall be at a volume of 1050 cc. ± 2 percent, when tested as specified in 4.4.2.1.3. All four sides of the bottle shall have molded

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in raised letters, 1/4 inch minimum in size, "For External Use Only", as shown on figure 1. On two opposite sides, the phrase "Capacity—1000 cc." shall be molded in raised letters, 3/16 inch minimum in size, near the bottom of the bottle. On the sides upon which the capacity is marked and at a mark corresponding to the 1050 cc. level, shall be a raised line, 1/16 inch wide and no less than 1½ inches long, which shall represent the solution level when filled. Under this line, in molded raised letters, 3/16 inch minimum in size, shall be the phrase "Fill to Here".

3.2.2 Closure cap. The closure cap for both sizes of the bottles shall be constructed in accordance with figure 3. The cap shall automatically seal itself to the rubber collar so as to retain a vacuum of 29½ inches of mercury after sterilization, as specified in 4.4.1.2.1.1, when the bottle is filled to the mark with solution.

3.2.3 Collar. The automatic closure collars for both sizes of the bottles shall be in accordance with figure 4. The inside of the collar shall have a groove, shaped as shown in figure 4, to grip the beaded neck of the bottle in order to retain the vacuum specified in 3.2.2.

3.3 Finish.

3.3.1 Bottle. Glass shall be properly distributed and well annealed. Glass shall be practically free of defects such as rough spots, blowovers, bubbles, and blisters.

3.3.2 Collar and cap. All edges shall be free from sharp, burred, and ragged edges. The cap and collar shall be uniformly colored. The cap and collar shall be free of matting, cracks, bubbles, and pitting. Collar and cap shall be symmetrically formed.

3.4 Identification marking. The name or registered trademark of the manufacturer shall be molded into the collar in raised letters of appropriate size, and shall appear on the top of the plastic cap.

3.5 Workmanship. Workmanship shall be first class throughout. The sterile fluid storage bottles shall be free from defects which detract from their appearance or may impair their serviceability.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major A defects	II	1.0
Total A (Major A and Minor A combined) defects	II	4.0
Major B defects	S-1	1.0
For dimensional examination	S-2	2.5

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4.2.2 For tests. Sampling for end item testing shall be conducted in accordance with MIL-STD-105.

Component	Characteristic	Requirement	Test procedure	Inspection level *
Bottle	Annealing	3.1.1	4.4.2.1.1	S-2
	Bursting pressure	3.1.1	4.4.2.1.2	S-2
	Capacity:			
	Size I bottle	3.2.1.1	4.4.2.1.3	S-1
	Size II bottle	3.2.1.2	4.4.2.1.3	S-1
	Vacuum retention	3.2.2	4.4.1.2.1.1	S-2
Cap	Autoclaving	3.1.2	4.4.2.2	S-2
Collar	Hardness	3.1.3	4.4.2.3	S-2

* Acceptance number shall be zero (0) for all sample sizes, regardless of lot size.

4.2.3 For material testing. Sufficient quantity of glass and plastic, prepared in conformance with test methods listed in 4.4.1 shall be selected for test. When sample size for components covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated for inspection level S-1 in MIL-STD-105. No test result shall fall below the minimum or exceed the maximum values specified. The sample unit for testing shall be the quantity necessary to perform all required tests one time each. Not less than three samples shall be taken regardless of lot size.

4.2.4 When possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The bottles shall be examined to determine compliance with all requirements contained in this specification.

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4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects *

Major

101 Bottle and cap not free from cracks, breaks, chips, sharp edges, and deformed areas.
102 Collar not free from cuts, tears, ruptures, and holes.
103 Bottle not stable; rocks on flat surface.
104 Collar fails to fit readily into plastic cap.
105 Cap fails to seat and unseat properly over collar mounted on the mouth of bottle.

Minor

201 Bottle not substantially free from seeds, striae, and bubbles.
202 Raised marking on bottle not present or complete.
203 Marking on cap not present and legible.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The bottles shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Material testing.

4.4.1.1 Glass. Glass used in the construction of the bottles shall be tested for the physical properties listed, in accordance with test methods of Fed. Test Method Std. No. 626, as follows:

Chemical durability ------ method 320 (method P-W)
Coefficient of expansion-- method 240 or 241

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4.4.1.2 Plastic. The plastic molding material used in the construction of the cap shall be tested for the properties listed, in accordance with the test methods of Fed. Test Method Std. No. 406, as follows:

Water absorption ---------- method 7031
Heat distortion temperature -------- method 2011.1

4.4.1.2.1 Deterioration tests.

4.4.1.2.1.1 Autoclaving. Plastic specimens and caps shall be autoclaved at 250°F. (15-17 pounds per square inch (p.s.i.)) for ten 30-minute periods, with a 10 minute rest period between each autoclaving.

4.4.1.2.1.2 Chemical resistance. Samples of plastic cap material shall be tested as specified in method 7011 (salt solution immersion) of Fed. Test Method Std. No. 406.

4.4.2 End item testing.

4.4.2.1 Bottle.

4.4.2.1.1 Annealing. Bottle shall be viewed in a polariscope to determine compliance with requirements for annealing, as specified in method 420 of Fed. Test Method Std. No. 626.

4.4.2.1.2 Bursting pressure. Each sample shall be completely abraded on its external surface, using 150-grit abrasive. External hydrostatic pressure shall be applied at a rate of 5 pounds per second to the sealed sample. The test shall be conducted at room temperature.

4.4.2.1.3 Capacity. Capacity, using distilled water, shall be determined accurately to the nearest cc.

4.4.2.2 Cap. Cap shall be autoclaved as specified in 4.4.1.2.1.1.

4.4.2.3 Collar. The hardness of the collar shall be tested in accordance with method 3021 of Fed. Test Method Std. No. 601. The rubber shall be tested unaged and after aging for 10 cycles of 30 minute duration in an autoclave at 250° F. (15-20 p.s.i.), with a 10-minute rest period between each autoclaving.

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5. PREPARATION FOR DELIVERY

(For civil agency procurement Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment.)

5.1 Packaging.

5.1.1 Unit of issue. Six (6) bottles or twelve (12) bottles as specified, constitute one unit.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit shall be packaged in a regular, slotted, single-wall, corrugated, fiberboard box of not less than 200-pound Mullen test, with "A" or "C" flute corrugations, meeting all requirements of PPP-B-636. The interior of each box shall be fitted with full-size top and bottom pads and three full-height scored shells. The shells shall be scored and folded to form two individual cells or one double shell. All corrugations shall be horizontal. Shells shall provide two thicknesses of combined fiberboard between adjacent glass containers, and one thickness between glass containers and the inside faces of the ends and sides of the box. Scored sheets and pads shall be fabricated of the same material as the box. Bottles shall fit snugly in the cells. Method I closure of PPP-B-636 shall be utilized.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit packaged as specified in 5.1.2.1 shall be overpacked in an exterior container designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

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5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. No overpacking shall be required.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. In addition, the following legend shall be shown:

"GLASS - HANDLE WITH CARE"

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

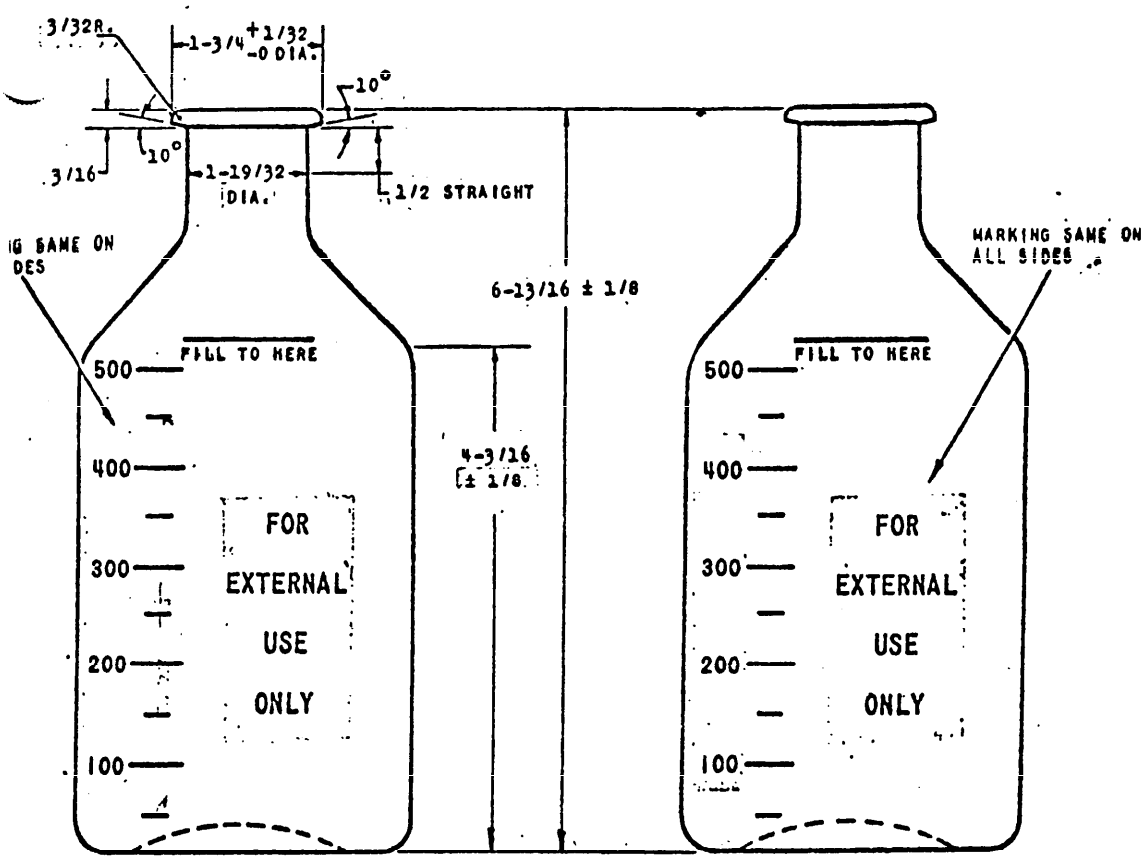
Military Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

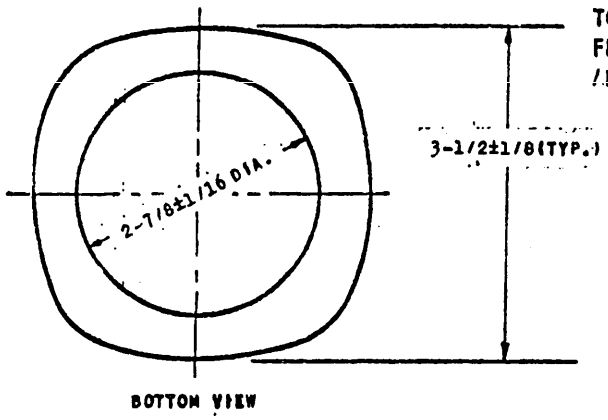
Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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ALL MEASUREMENTS ARE IN INCHES

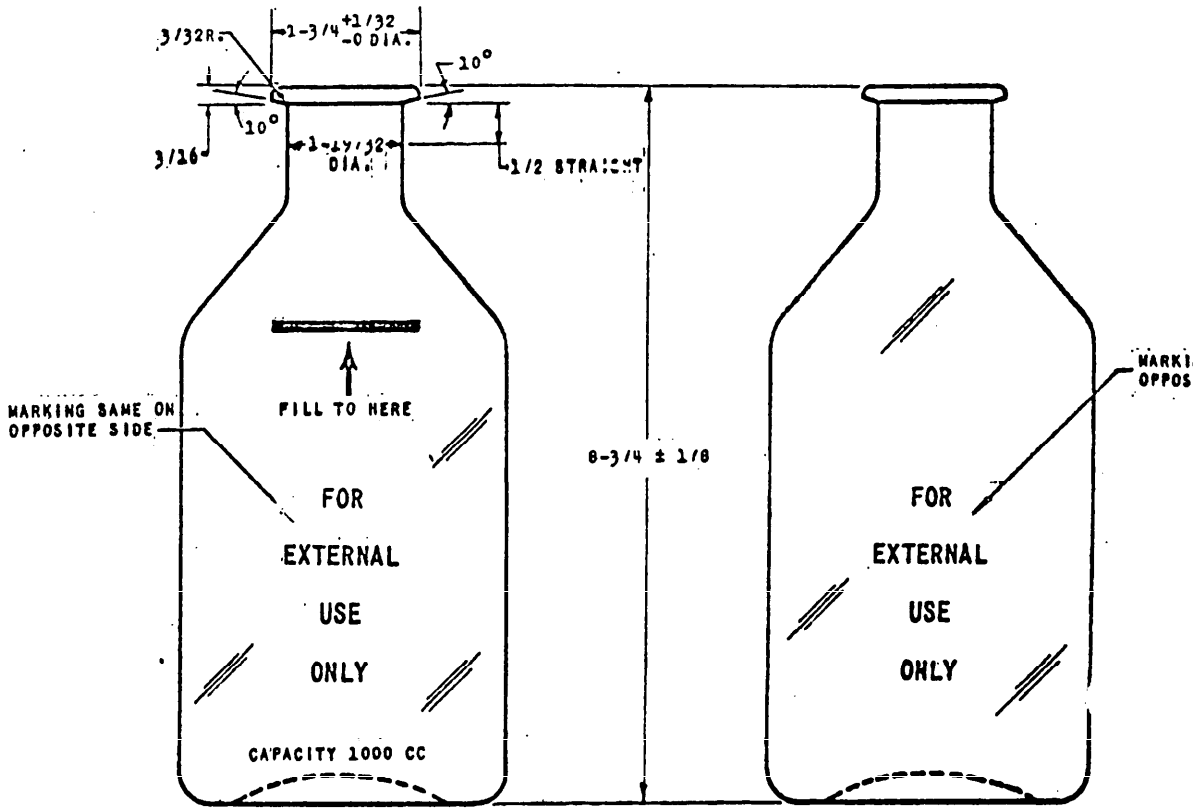


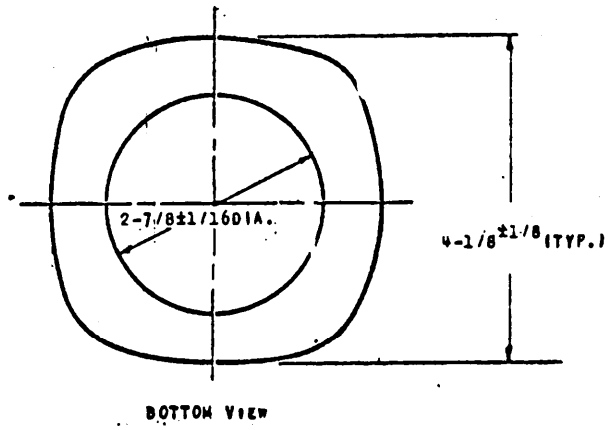
TOLERANCES UNLESS OTHERWISE SPECIFIED:
FRACTIONS = ±1/32
ANGLES = ±2°

BOTTLE, SIZE I

FIGURE 1

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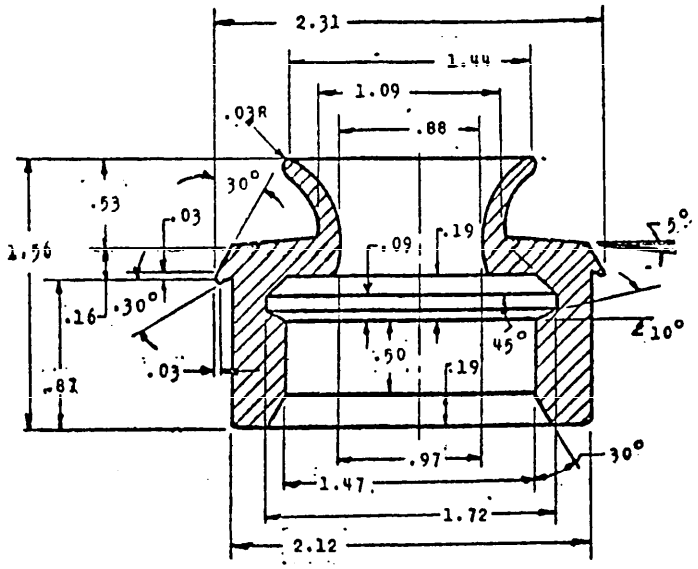


TOLERANCES
UNLESS OTHERWISE SPECIFIED:
FRACTIONS - ±1/32
ANGLES - ±2°

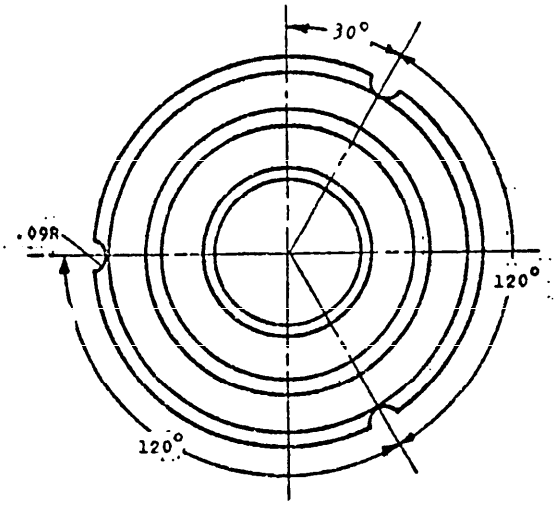
BOTTLE, SIZE 11

FIGURE 2

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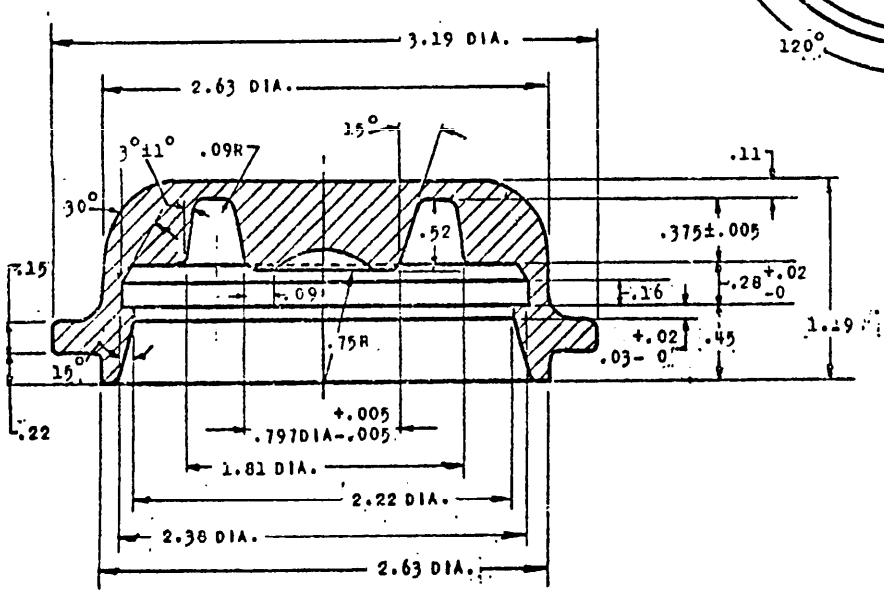


TOLERANCES
UNLESS OTHERWISE SPECIFIED:
DECIMAL = ± .01
ANGLES = ± 2°



RUBBER COLLAR NEOPRENE

FIGURE 4



CLOSURE CAP, PLASTIC

FIGURE 3

NOTE: ALL MEASUREMENTS ARE IN INCHES

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

DD-B-00592a(DSA-DM) BOTTLE, STERILE FLUID STORAGE 30 March 1965

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426. 3ND PPSO 72302

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

FOLD UNDER

DEFENSE SUPPLY AGENCY
DEFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

POSTAGE FEES PAID
DEFENSE SUPPLY AGENCY

OFFICIAL BUSINESS

DEFENSE SUPPLY AGENCY

DEFENSE MEDICAL SUPPLY CENTER

ATTN: DIRECTORATE OF TECHNICAL OPERATIONS (CODE T-15)

3RD AVENUE AND 29TH STREET

BROOKLYN, N.Y., 11232

FOLD

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